

CSE 309 Theory of Automata, Spring 2026
Exercise 8

1. **A Big Oh-No.** Answer whether True or False with brief reasoning.

- (a) $2n = O(n)$
- (b) $n^2 = O(n)$
- (c) $n^2 = O(n \log^2 n)$
- (d) $n \log n = O(n^2)$
- (e) $3^n = 2^{O(n)}$
- (f) $2^{2^n} = O(2^{2^n})$

2. **Polynomial Closure.** Show that the class P is closed under union and complement.

3. **Finding Triangles.** A *triangle* in an undirected graph G is a 3-clique. Show that $TRIANGLE \in P$, where

$$TRIANGLE = \{\langle G \rangle \mid G \text{ contains a triangle}\}.$$

4. In both parts, provide an analysis of the time complexity of your algorithm.

- (a) Show that $EQUAL_{DFA} \in P$.
- (b) Say that a language A is *star-closed* if $A = A^*$. Give a polynomial time algorithm to test whether a DFA recognizes a star-closed language. (Note that $EQUAL_{NFA}$ is not known to be in P .)